

Finite Automaton with output

Finite automaton discussed so far, is just associated with the RE or the language.

There is a question whether does there exist an FA which generates an output string corresponding to each input string ? The answer is yes. Such machines are called machines with output.

There are two types of machines with output. Moore machine and Mealy machine

Moore machine

A Moore machine consists of the following

1. A finite set of states $q_0, q_1, q_2, \dots$ where q_0 is the initial state.
2. An alphabet of letters $\Sigma = \{a, b, c, \dots\}$ from which the input strings are formed.
3. An alphabet $\Delta = \{x, y, z, \dots\}$ of output characters from which output strings are generated.

Moore machine continued ...

4. A transition table that shows for each state and each input letter what state is entered the next.
5. An output table that shows what character is printed by each state as it is entered.

Moore machine continued ...

Note: It is to be noted that since in Moore machine no state is designated to be a final state, so there is no question of accepting any language by Moore machine. However in some cases the relation between an input string and the corresponding output string may be identified by the Moore machine. Moreover, the state to be initial is not important as if the machine is used several times and is restarted after some time, the machine will be started from the state where it was left off. Following are the examples

Example

Consider the following Moore machine having the states q_0, q_1, q_2, q_3 where q_0 is the start state and

$$\Sigma = \{a, b\},$$

$$\square = \{0, 1\}$$

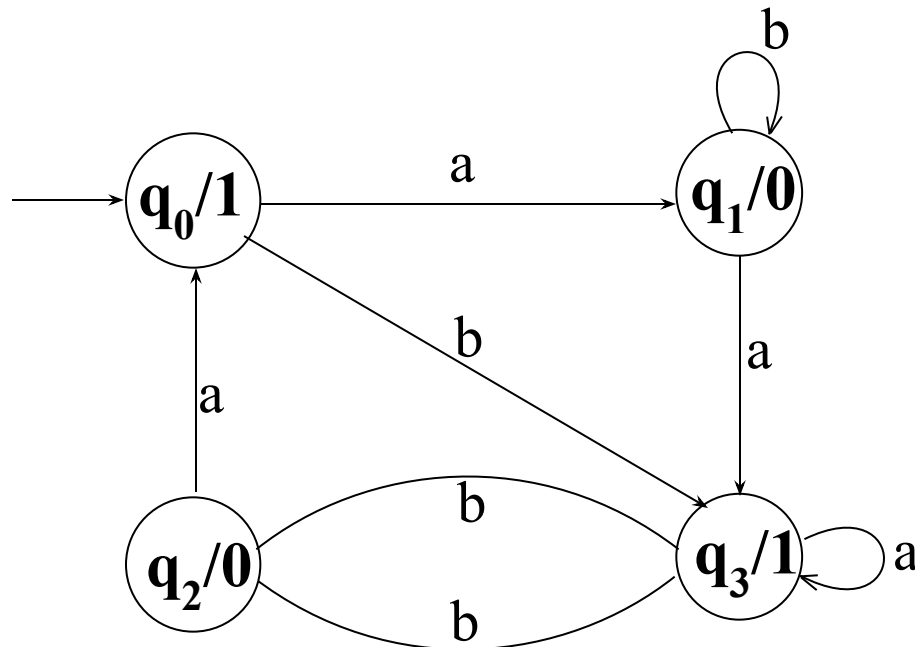
the transition table follows as

Example continued ...

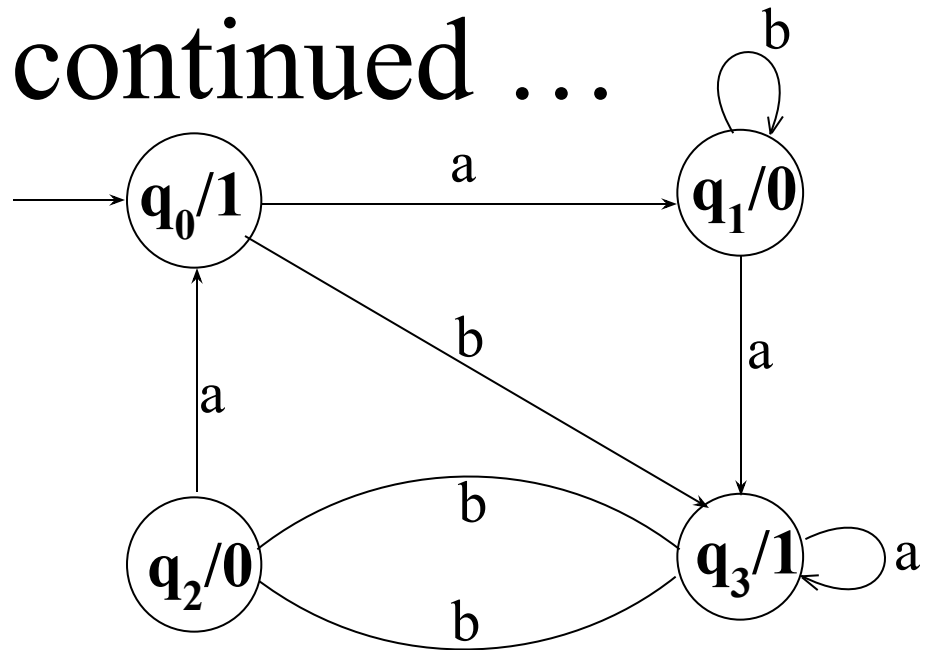
Old States	New States after reading		Characters to be printed
	a	b	
q_0^-	q_1	q_3	1
q_1	q_3	q_1	0
q_2	q_0	q_3	0
q_3	q_3	q_2	1

Example continued ...

the transition diagram corresponding to the previous transition table may be

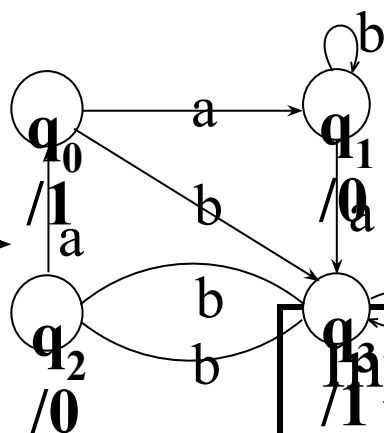


Example continued ...



It is to be noted that the states are labeled along with the characters to be printed.

Running the string `abbabbba` over the above machine, the corresponding output string will be `100010101`, which can be determined by the following table as well



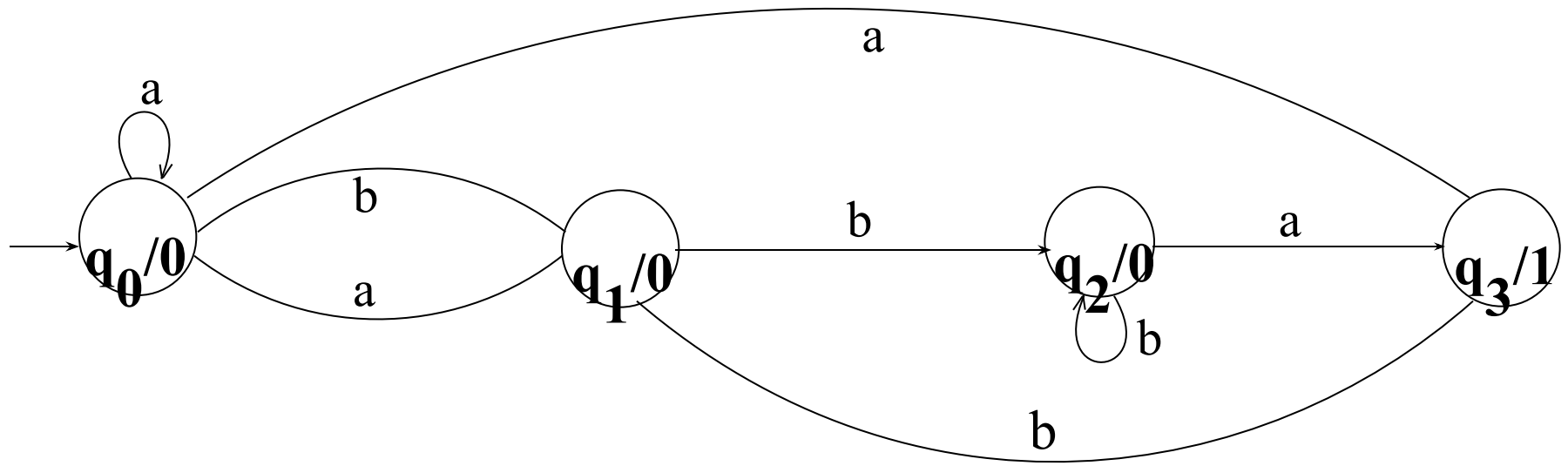
Example continued ...

Input		a	b	b	a	b	b	b	a
State	q_0	q_1	q_1	q_1	q_3	q_2	q_3	q_2	q_0
output	1	0	0	0	1	0	1	0	1

It may be noted that the length of output string is 1 more than that of input string as the initial state prints out the extra character 1, before the input string is read.

Example

To identify the relation between the input strings and the corresponding output strings in the following Moore machine,



Example continued ...

if the string bbbabaabbaa is run, the output string will be 000010000010, as shown below

Input		b	b	b	a	b	a	a	b	b	a	a
State	q_0	q_1	q_2	q_2	q_3	q_1	q_0	q_0	q_1	q_2	q_3	q_0
output	0	0	0	0	1	0	0	0	0	0	1	0

Example continued ...

It can be observed from the given Moore machine that q_3 is the only state which prints out the character 1 which shows that the moment the state q_3 is entered, the machine will print out 1. To enter the state q_3 , starting from q_0 the string must contain bba. It can also be observed that to enter the state q_3 once more the string must contain another substring bba. In general the input string will visit the state q_3 as many times as the number of substring bba occurs in the input string. Thus the number of 1's in an output string will be same as the number of substring bba occurs in the corresponding input string.

Mealy machine

A Mealy machine consists of the following

1. A finite set of states $q_0, q_1, q_2, \dots$ where q_0 is the initial state.
2. An alphabet of letters $\Sigma = \{a, b, c, \dots\}$ from which the input strings are formed.
3. An alphabet $\Delta = \{x, y, z, \dots\}$ of output characters from which output strings are generated.

Mealy machine continued ...

4. A pictorial representation with states and directed edges labeled by an input letter along with an output character. The directed edges also show how to go from one state to another corresponding to every possible input letter.

Note: It is not possible to give transition table in this case.

Mealy machine continued ...

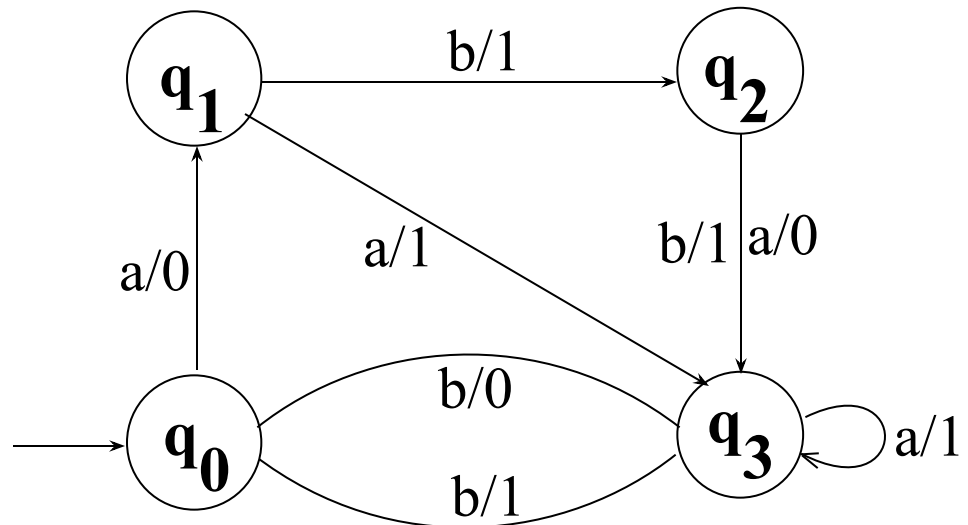
Note: It is to be noted that since, similar to Moore machine, in Mealy machine no state is designated to be a final state, so there is no question of accepting any language by Mealy machine. However in some cases the relation between an input string and the corresponding output string may be identified by the Mealy machine. Moreover, the state to be initial is not important as if the machine is used several times and is restarted after some time, the machine will be started from the state where it was left off. Following are the examples

Example

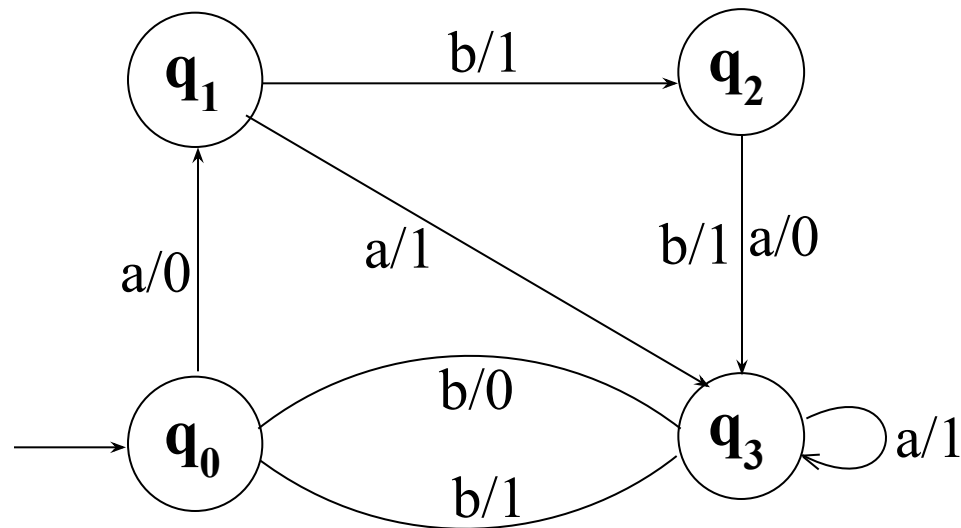
Consider the following Mealy machine having the states q_0, q_1, q_2, q_3 , where q_0 is the start state and

$$\Sigma = \{a, b\},$$

$$\square = \{0, 1\}$$



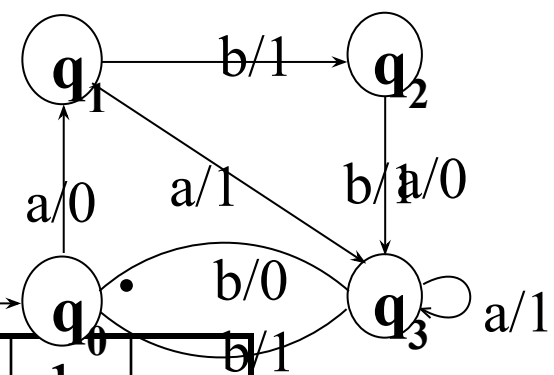
Example continued ...



Running the string abbabbba over the above machine, the corresponding output string will be 11011010, which can be determined by the following table as well

Example continued

Input		a	b	b	a	b	b	b	a
States	q_0	q_1	q_2	q_3	q_3	q_0	q_3	q_0	q_1
output		0	1	1	1	1	0	1	0



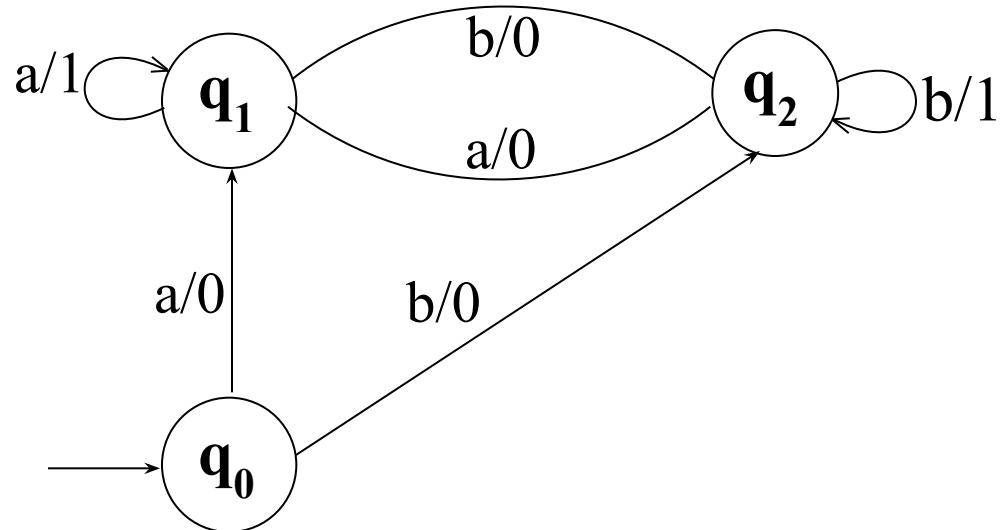
It may be noted that in Mealy machine, the length of output string is equal to that of input string.

Example

Consider the following Mealy machine having the states q_0 , q_1 , q_2 , where q_0 is the start state and

$$\Sigma = \{a,b\},$$

$$\square = \{0,1\}$$



Example continued ...

It is observed that in the above Mealy machine, if in the output string the n th character is 1, it shows that the n th letter in the input string is the second in the pair of double letter.

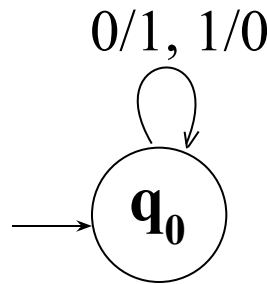
For babaababba as input string the machine will print 0000100010.

Example

Consider the following Mealy machine having the only state q_0 as the start state and

$$\Sigma = \{0,1\},$$

$$\square = \{0,1\}$$



If 0011010 is run on this machine then the corresponding output string will be 1100101. This machine is also called **Complementing machine**.

Constructing the incrementing machine

In the previous example of complementing machine, it has been observed that the input string and the corresponding output string are 1's complement of each other. There is a question whether the Mealy machine can be constructed, so that the output string is increased, in magnitude, by 1 than the corresponding input string ? The answer is yes. This machine is called the incrementing machine. Following is how to construct the incrementing machine

Constructing the incrementing machine continued ...

Before the incrementing machine is constructed, consider how 1 is added to a binary number.

Since, if two numbers are added, the addition is performed from right to left, so while increasing the binary number by 1, the string (binary number) must be read by the corresponding Mealy machine from right to left, and hence the output string (binary number) will also be generated from right to left.

Constructing the incrementing machine continued ...

Consider the following additions

$$\text{a) } 100101110 \quad \text{b) } 1001100111$$

$$+ 1$$

$$+ 1$$

$$\underline{100101111}$$

$$\underline{1001101000}$$

It may be observed from the above that

- a) If the right most bit of binary number, to be incremented, is 0, the output binary number can be obtained by converting the right most bit to 1 and remaining bits unchanged.

Constructing the incrementing machine continued ...

b) If the right most bit of binary number is 1 then the output can be obtained, converting that 1 along with all its concatenated 1's to 0's, then converting the next 0 to 1 and remaining bits unchanged.

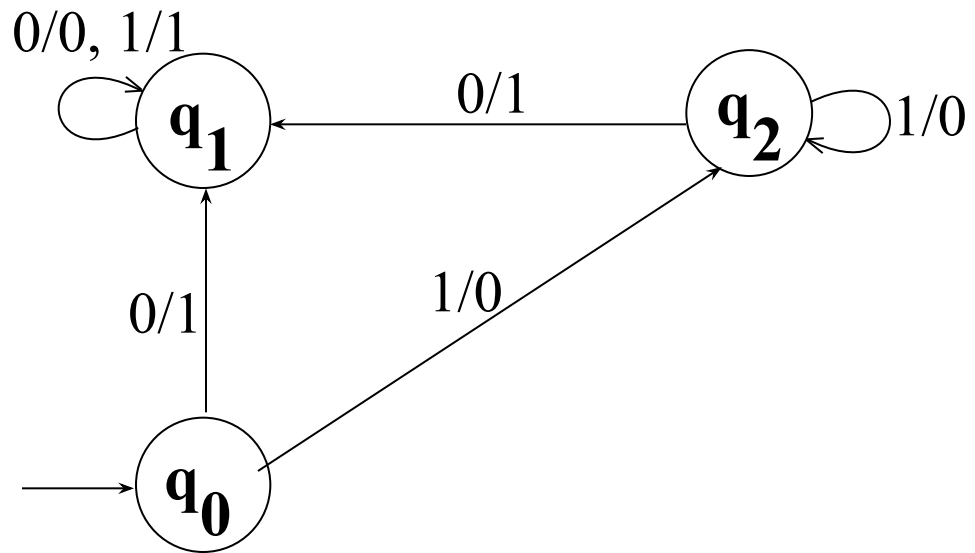
The observations (a) and (b) help to construct the following Incrementing (Mealy) machine.

Constructing the incrementing machine continued ...

The Mealy machine have the states

q_0, q_1, q_2 , where q_0 is the start state and $\Sigma = \{0,1\}$,

$\square = \{0,1\}$



Constructing the incrementing machine continued ...

It may be observed that, in the incrementing machine, if 0 is read at initial state q_0 , that 0 is converted to 1 and a no change state q_1 (no carry state) is entered where all 0's and all 1's remain unchanged. If 1 is read at initial state, that 1 is converted to 0 and the state q_2 (owe carry state) is entered, where all 1's are converted to 0's and at that state if 0 is read that 0 is converted to 1 and the machine goes to no change state.

If the strings 100101110 and 1001100111 are run over this machine, the corresponding output strings will be 100101111 and 1001101000 respectively.

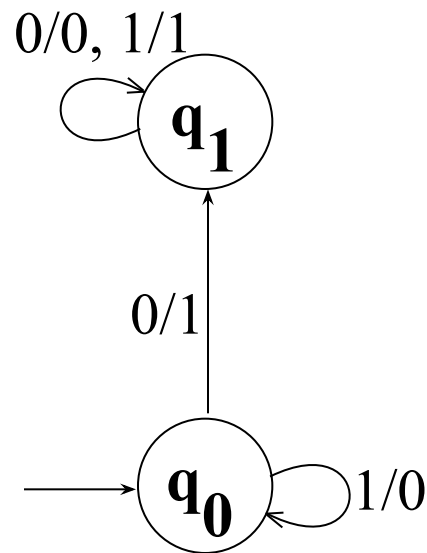
Note

It is to be noted that if the string 111111 is run over the incrementing machine, the machine will print out 000000, which is not increased in magnitude by 1. Such a situation is called an overflow situation, as the length of output string will be same as that of input string.

It may also be noted that there exists another incrementing machine with two states.

Solution of the Task

Incrementing machine with two states



Applications of Incrementing and Complementing machines

1's complementing and incrementing machines which are basically Mealy machines are very much helpful in computing.

The incrementing machine helps in building a machine that can perform the addition of binary numbers.

Using the complementing machine along with incrementing machine, one can build a machine that can perform the subtraction of binary numbers, as shown in the following method

Subtracting a binary number from another

Method: To subtract a binary b from a binary number a

1. Add 1's complement of b to a (ignoring the overflow, if any)
2. Increase the result, in magnitude, by 1 (use the incrementing machine). Ignoring the overflow if any.

Note: If there is no overflow in (1). Take 1's complement once again in (2), instead. This situation occurs when b is greater than a , in magnitude. Following is an example of subtraction of binary numbers³¹

Example

To subtract the binary number 101 from the binary number 1110, let

$a = 1110$ and $b = 101 = 0101$.

(Here the number of digits of b are equated with that of a)

i) Adding 1's complement (1010) of b to a .

1110

+1010

11000 which gives 1000 (ignoring the overflow)

Example continued ...

ii) Using the incrementing machine, increase the above result 1000, in magnitude, by 1

$$\begin{array}{r} 1000 \\ +1 \\ \hline \end{array}$$

1001 which is the same as obtained by ordinary subtraction.

Note

It may be noted that the above method of subtraction of binary numbers may be applied to subtraction of decimal numbers with the change that 9's complement of b will be added to a , instead in step (1). Following is the task in this regard

Task

Subtract 39 from 64

Solution: Taking $a=64$ and $b=39$.

i) Adding 9's complement (60) of b to a .

64

+60

124 which gives 24 (ignoring the overflow)

ii) Increasing the above result 24, in magnitude, by 1

+1

25 which is the same as obtained by ordinary subtraction.

Equivalent machines

Two machines are said to be **equivalent** if they print the same output string when the same input string is run on them.

Remark: Two Moore machines may be equivalent. Similarly two Mealy machines may also be equivalent, but a Moore machine can't be equivalent to any Mealy machine. However, ignoring the extra character printed by the Moore machine, there exists a Mealy machine which is equivalent to the Moore machine.

Theorem

Statement:

For every Moore machine there is a Mealy machine that is equivalent to it (ignoring the extra character printed by the Moore machine).

Proof: Let M be a Moore machine, then shifting the output characters corresponding to each state to the labels of corresponding incoming transitions, machine thus obtained will be a Mealy machine equivalent to M .

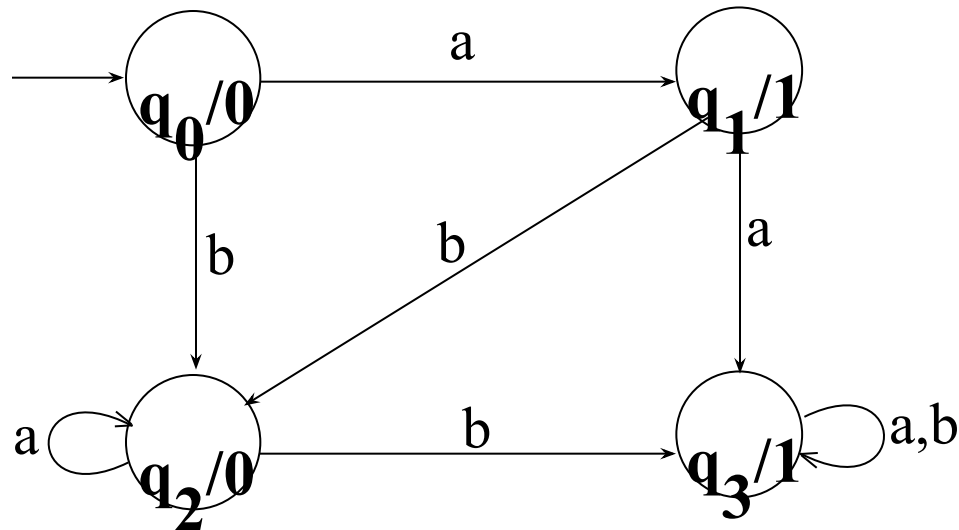
Following is a note

Note

It may be noted that while converting a Moore machine into an equivalent Mealy machine, the output character of a state will be ignored if there is no incoming transition at that state. A loop at a state is also supposed to be an incoming transition. Following is the example of converting a Moore machine into an equivalent Mealy machine

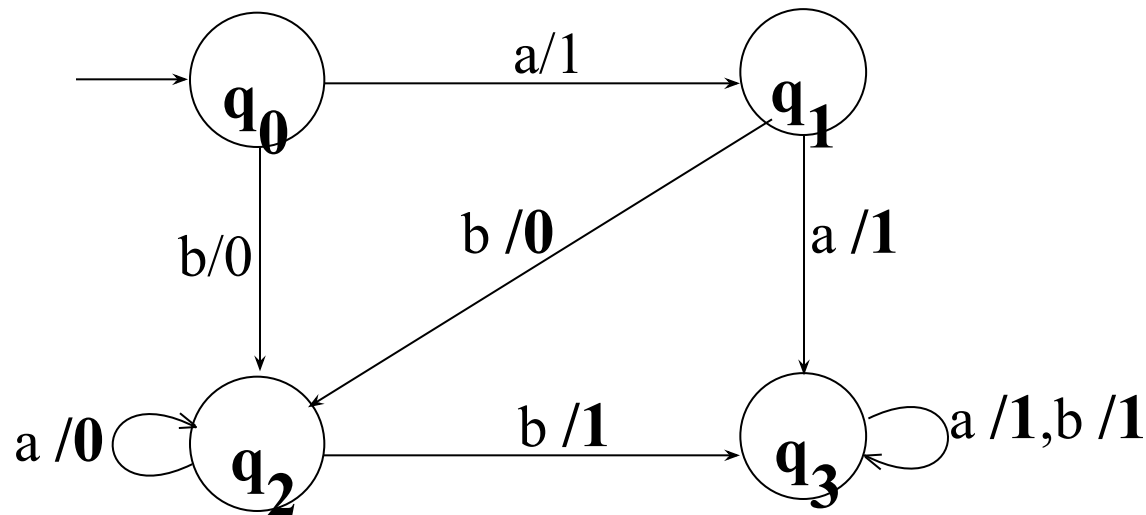
Example

Consider the following Moore machine



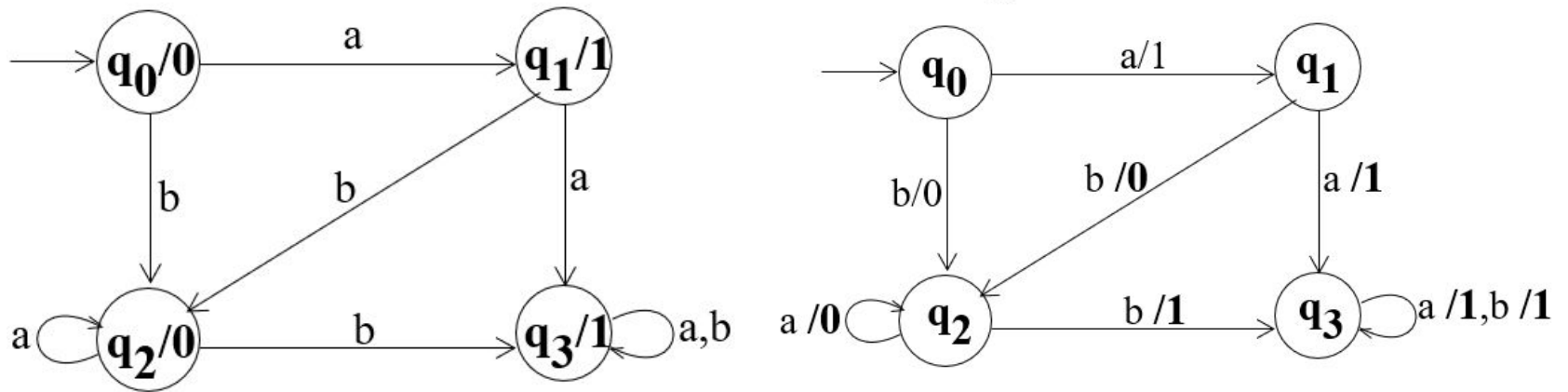
Using the method described earlier, the above machine may be equivalent to the following Mealy machine

Example continued ...



Running the string abbabbba on both the machines, the output string can be determined by the following table

Example continued ...



Input		a	b	b	a	b	b	b	a
States	q_0	q_1	q_2	q_3	q_3	q_3	q_3	q_3	q_3
Moore	0	1	0	1	1	1	1	1	1
Mealy		1	0	1	1	1	1	1	1

Theorem

Statement:

For every Mealy machine there is a Moore machine that is equivalent to it (ignoring the extra character printed the Moore machine).

Proof: Let M be a Mealy machine. At each state there are two possibilities for incoming transitions

1. The incoming transitions have the same output character.
2. The incoming transitions have different output characters.

Proof continued ...

If all the transitions have same output characters, then shift that character to the corresponding state.

If all the transitions have different output characters, then the state will be converted to as many states as the number of different output characters for these transitions, which shows that if this happens at state q_i then q_i will be converted to q_i^1 and q_i^2 *i.e.* if at q_i there are the transitions with two output characters then q_i^1 for one character and q_i^2 for other character.

Proof continued ...

Shift the output characters of the transitions to the corresponding new states q_i^1 and q_i^2 . Moreover, these new states q_i^1 and q_i^2 should behave like q_i as well. Continuing the process, the machine thus obtained, will be a Moore machine equivalent to Mealy machine M .

Following is a note

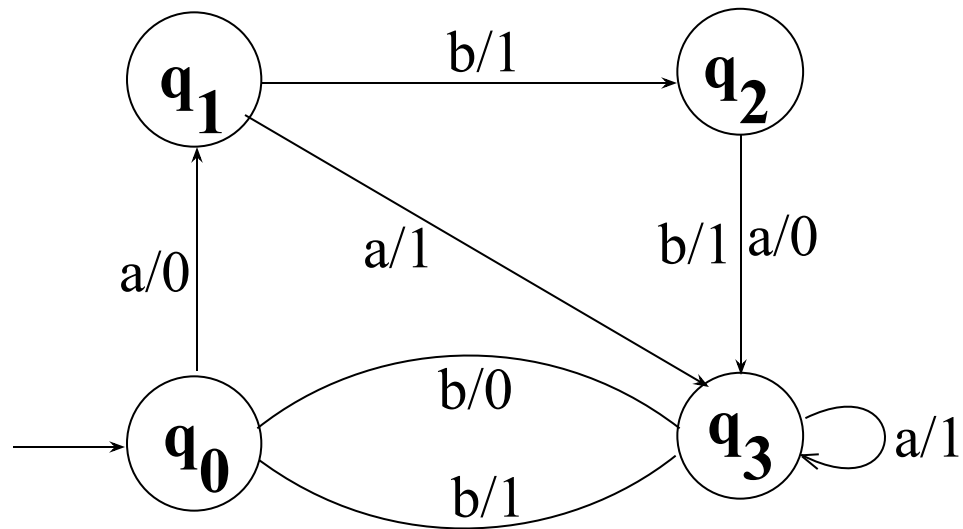
Note

It may be noted that if there is no incoming transition at certain state then any of the output characters may be associated with that state.

It may also be noted that if the initial state is converted into more than one new states then only one of these new states will be considered to be the initial state. Following is an example

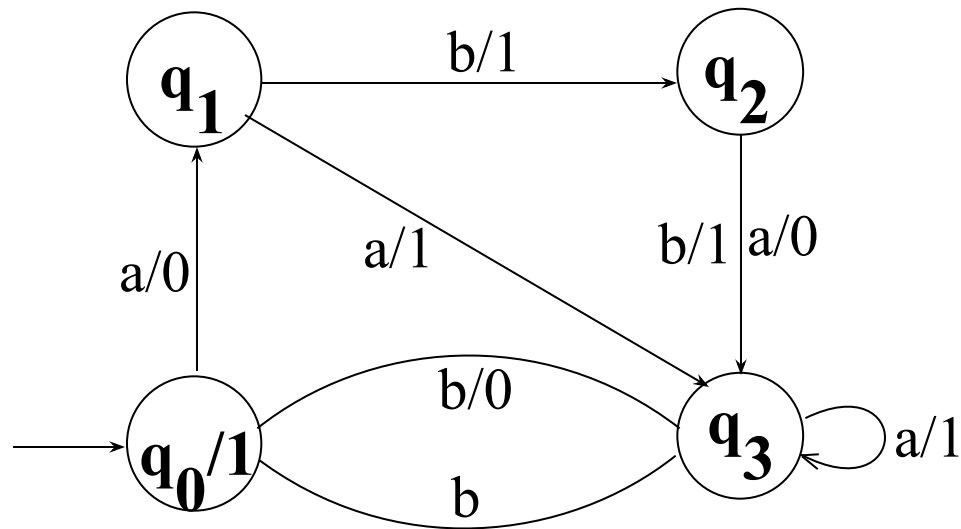
Example

Consider the following Mealy machine



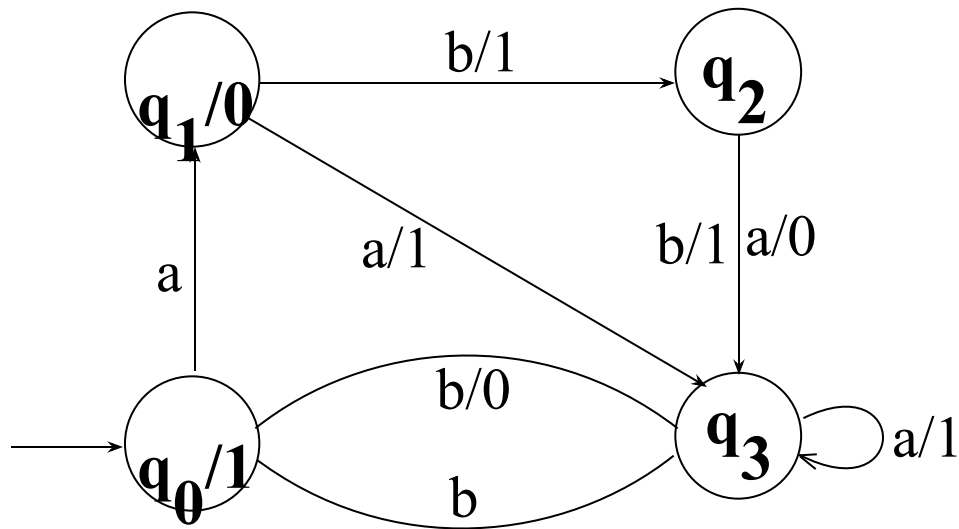
Example continued ...

Shifting the output character 1 of transition b to q_0



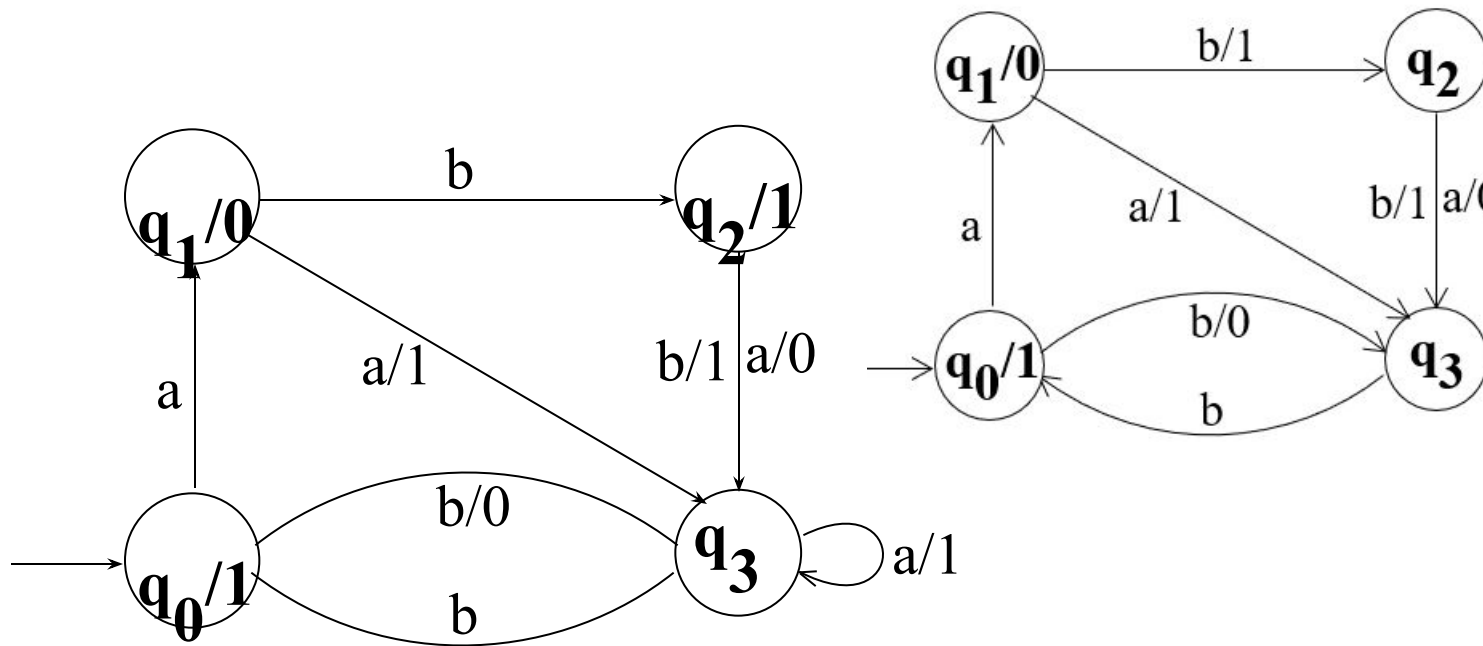
Example continued ...

Shifting the output character 0 of transition a to q_1



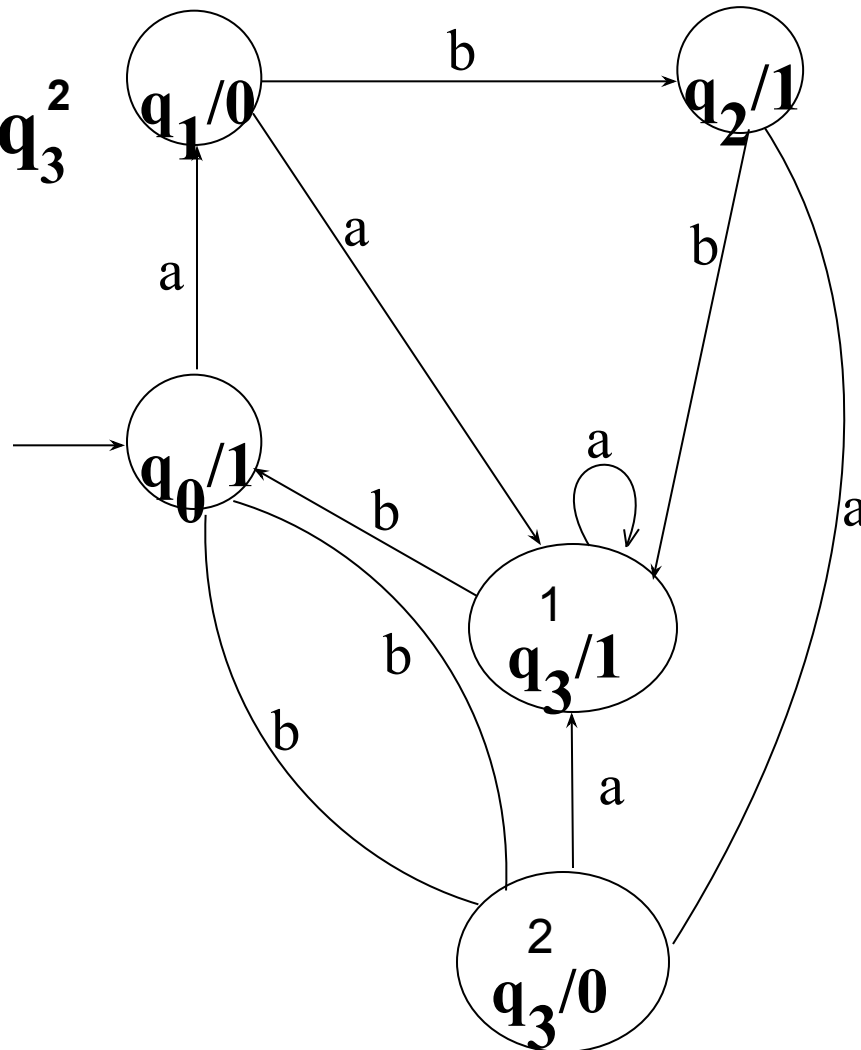
Example continued ...

Shifting the output character 1 of transition b to **a**.



Example continued ...

Splitting q_3
into q_3^1 and q_3^2

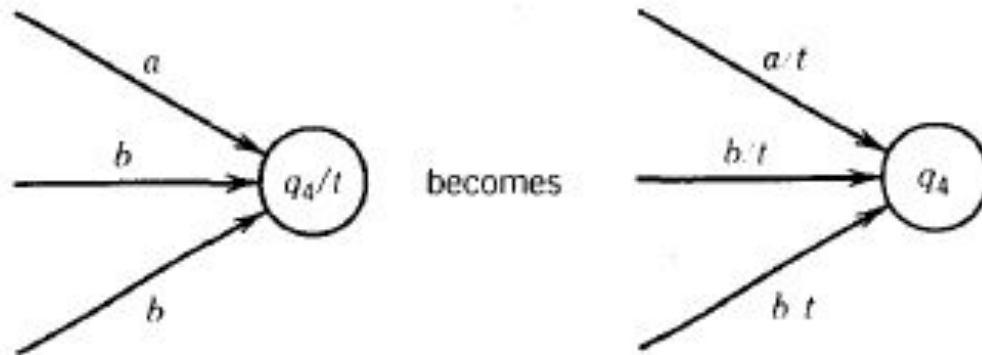


Theorem

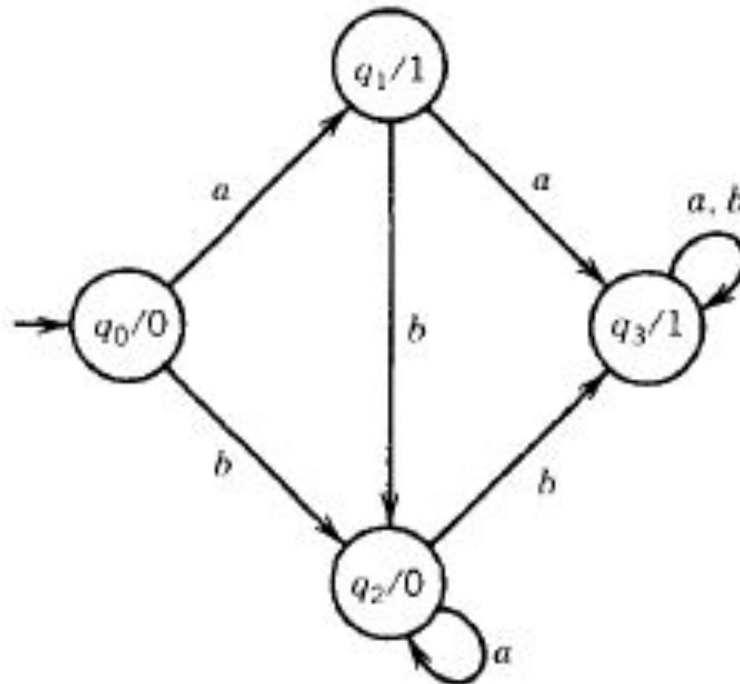
Statement:

For every Mealy machine there is a Moore machine that is equivalent to it (ignoring the extra character printed the Moore machine).

Steps



EXAMPLE



Solution

